# Synthetic data quality by document length

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

| Group | $n_D$ | $n_F$ | Method | Dilution and relevance | | | Fragmentation | |
|---|---|---|---|---|---|---|---|---|
| | | | | No-support ↓ | Query focus ↑ | Support recall ↑ | Joint exercise ↑ | Joint case ↑ |
| 8K | 249 | 60 | Vanilla Cartridges | 65.35 | 2.48 | 81.58 | 0.00 | 0.00 |
| | 249 | 60 | Question-guided (ours) | **35.53** | **99.28** | **92.57** | **32.14** | **86.67** |
| 16K | 250 | 71 | Vanilla Cartridges | 83.57 | 2.03 | 64.49 | 0.00 | 0.00 |
| | 250 | 71 | Question-guided (ours) | **44.83** | **98.57** | **95.20** | **20.61** | **81.69** |
| 32K | 247 | 81 | Vanilla Cartridges | 91.13 | 1.22 | 40.16 | 0.00 | 0.00 |
| | 247 | 81 | Question-guided (ours) | **51.23** | **97.58** | **91.70** | **22.05** | **79.01** |
| 64K | 247 | 84 | Vanilla Cartridges | 95.23 | 1.34 | 22.65 | 0.00 | 0.00 |
| | 247 | 84 | Question-guided (ours) | **55.29** | **97.28** | **84.72** | **21.71** | **70.24** |

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.